SpendWise: Modern MERN Expense Tracker

Full Stack Frameworks – Mini Project Evaluation

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Project Scope & Objectives

- Full CRUD daily income and expense tracking with analytics.
- MongoDB aggregation engine computing real-time monthly totals.
- Built with modern **MERN** stack (MongoDB, Express, React, Node).

Live Evaluation Links

- **Live Cloud App:** mern-expense-tracker-jskq.onrender.com
- **Cloud Database:** MongoDB Atlas (M0 Cluster0 on AWS)
- **GitHub Repository:** github.com/aksharsakhi/mern-expense-tracker

Architecture, REST APIs & MongoDB Aggregations

Three-Tier System Architecture

- **Frontend (React 18 + Vite):** Glassmorphism UI, Chart.js visuals.
- **Backend (Node.js + Express):** RESTful API, CORS, input validators.
- **Database (MongoDB 8.x):** Compound indexes on `(date, type)`.

Multi-Faceted MongoDB Aggregation

Single round-trip database query utilizing `$facet`:

- **Totals Facet:** Computes total income, expenses, count.
- **Category Distribution:** Group expenditures to calculate percentage share per category for donut chart.
- **Daily Cashflow Velocity:** Group daily spending to feed dynamic bar chart.

RESTful Endpoints Contract

- `GET /api/transactions?month=2026-10` (Filtered list)
- `GET /api/transactions/summary` (MongoDB aggregation)
- `POST /api/transactions` (Create record with validation)
- `PUT /api/transactions/:id` & `DELETE` (Update/Remove)
- `POST /api/transactions/seed` (One-click demo dataset)

Computed Real-Time Metrics

- **Net Balance** = Total Income − Total Expense
- **Savings Rate** = $\frac{\text{Income} - \text{Expense}}{\text{Income}} \times 100\%$
- **Payment Splits:** Breakdown across UPI, Cards, Net Banking.

Evaluation Demo, Deployment & Key Takeaways

Live Demonstration Workflow

- **Dynamic Month Switching:** Instant metric recalculation between October and September 2026.
- **Full CRUD Operations:** Add new transactions via modal, instant validation, update notes, delete with confirmation.
- **Multi-Criteria Filters:** Live text search + type tabs (`All`, `Income`, `Expense`) + category dropdowns.
- **Data Portability:** Direct one-click CSV export of monthly records.

Evaluation Learning Outcomes

- Production-grade full-stack MERN engineering.
- Scalable MongoDB aggregation optimization.
- Accessible, aesthetic UI with fluid Chart.js data visualization.

DevOps & Production Deployment

- **Render Cloud:** Deployed 24/7 web service with automatic CI/CD.
- **Database Cloud:** MongoDB Atlas with multi-region replica sets.
- **Unified Server:** Express serves compiled React bundle (`client/dist`) in production.
- **Docker Ecosystem:** Multi-stage container builds + `docker-compose.yml` with Nginx reverse proxy.
- **Cloud Blueprints:** Preconfigured `render.yaml` and `vercel.json` for cloud platform deployments.

Live Demonstration Link

https://mern-expense-tracker-jskq.onrender.com